[MP2] MP2_MIN_AVAIL_SLAB and Printing struct kmem_cache Information (Solved)

林秉翰 (LIN, BING-HAN)

All Sections

5 5

助教您好：

由於測試mp2-7時會機率性報錯[Invalid print message:{.....}is broken or not in partial]，經多次驗證，報錯原因與race condition有關，先後執行free object與執行print_kmem_cache()導致結果不一致。具體報錯皆發生在印出partial slab list中包含1個半滿的slab + 1個全空的slab (目前沒有實作free slab list所以空的slab會回到partial中，且1+1尚未"exceed" MP2_MIN_AVAIL_SLAB)。

以下想請問助教：

1. “the number of (partial + free) slabs exceeds MP2_MIN_AVAIL_SLAB”應是指 `>MP2_MIN_AVAIL_SLAB` 而非 `>=MP2_MIN_AVAIL_SLAB` 。

2. 在沒有實作free slab的情況下，印出partial slab時可否接受印出個全空的slab？

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楊子慶 (YANG, TZU-CHING) (https://cool.ntu.edu.tw/courses/46296/users/229158)

Yesterday

同學您好：

1. 是的

2. 可以，助教的八種實作中其中四種便是沒有實作 free，空的直接放放在 partial 中。

助教，子慶

Reply

(http 林秉翰 (LIN, BING-HAN) (https://cool.ntu.edu.tw/courses/46296/users/188744)
Yesterday

感謝助教回覆！

基於上述二者實作下，機率性報錯[Invalid print message:{.....}is broken or not in partial]仍然發生，翻閱討論版發現也有同學遇到類似問題: [MP2] checker 評分標準與 spec 不一致 (https://cool.ntu.edu.tw/courses/46296/discussion_topics/385950?wrap=1) 。此篇留言助教提供的範例

// mytest.txt

echo Ok | gah . 10 | oak end
mp2

./mp2.sh test custom

經測試會是**FAIL**(100%)，輸出則與助教示範相似，結果如下：

```
❯ ./mp2.sh update
/usr/bin/curl
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100  4143  100  4143    0     0  19180      0 --:--:-- --:--:-- --:--:-- 19269
Your mp2 is the newest version!
❯ cat test/custom/mytest.txt
echo Ok | gah . 10 | oak end
mp2
❯ ./mp2.sh test custom
== Test custom/mytest (0%) ==
(2.5s)
    Invalid print message: {'addr': 2280300544, 'freelist': 2280303088, 'nxt': 0, 'objs': [{'addr': 22803
00568, 'as_ptr': 2280303592, 'as_obj': {'tp': -2014663704, 'ref': 0, 'readable': 1, 'writable': 1, 'pipe'
: 361700864190383365, 'ip': 2147702848, 'off': 0, 'major': 1285}}, {'addr': 2280301072, 'as_ptr': 2280300
568, 'as_obj': {'tp': -2014666728, 'ref': 0, 'readable': 1, 'writable': 1, 'pipe': 361700864190383365, 'i
p': 2147702984, 'off': 0, 'major': 1285}}, {'addr': 2280301576, 'as_ptr': 2280301072, 'as_obj': {'tp': -2
014666224, 'ref': 0, 'readable': 1, 'writable': 1, 'pipe': 361700864190383365, 'ip': 2147703120, 'off': 0
, 'major': 1285}}, {'addr': 2280302080, 'as_ptr': 2280301576, 'as_obj': {'tp': -2014665720, 'ref': 0, 're
adable': 1, 'writable': 1, 'pipe': 361700864190383365, 'ip': 2147703256, 'off': 0, 'major': 1285}}, {'add
r': 2280302584, 'as_ptr': 2280302080, 'as_obj': {'tp': -2014665216, 'ref': 0, 'readable': 1, 'writable':
1, 'pipe': 361700864190383365, 'ip': 2147703392, 'off': 0, 'major': 1285}}, {'addr': 2280303088, 'as_ptr'
: 2280302584, 'as_obj': {'tp': -2014664712, 'ref': 0, 'readable': 1, 'writable': 1, 'pipe': 3617008641903
83365, 'ip': 2147703528, 'off': 0, 'major': 1285}}, {'addr': 2280303592, 'as_ptr': 2280304096, 'as_obj':
{'tp': -2014663200, 'ref': 0, 'readable': 5, 'writable': 5, 'pipe': 361700864190383365, 'ip': 36170086419
0383365, 'off': 84215045, 'major': 1285}}, {'addr': 2280304096, 'as_ptr': 0, 'as_obj': {'tp': 0, 'ref': 0
, 'readable': 5, 'writable': 5, 'pipe': 361700864190383365, 'ip': 361700864190383365, 'off': 84215045, 'm
ajor': 1285}}]} is broken or not in partial
    QEMU output saved to out/custom/mytest.out.failed
Score: 0/0
```

```
108  [SLAB] Free 0x0000000087f4e210 in slab 0x0000000087f4e000 (file_cache)
109  [SLAB] End of free
110  $ mp2
111  [SLAB] kmem_cache { name: file_cache, object_size: 504, at: 0x0000000087f59000, in_cache_obj: 0 }
112  [SLAB]  [ partial slabs ]
113  [SLAB]      [ slab 0x0000000087f4e000 ] { freelist: 0x0000000087f4e210, nxt: 0x0000000087eaa000 }
114  [SLAB]          [ idx 0 ] { addr: 0x0000000087f4e018, as_ptr: 0x0000000900000003, as_obj: { tp: 3, ref: 9, readable: 1, wr
115  [SLAB]          [ idx 1 ] { addr: 0x0000000087f4e210, as_ptr: 0x0000000087f4ede0, as_obj: { tp: -2013991456, ref: 0, reada
116  [SLAB]          [ idx 2 ] { addr: 0x0000000087f4e408, as_ptr: 0x0000000087f4e7f8, as_obj: { tp: -2013992968, ref: 0, reada
117  [SLAB]          [ idx 3 ] { addr: 0x0000000087f4e600, as_ptr: 0x0000000000000000, as_obj: { tp: 0, ref: 0, readable: 1, wr
118  [SLAB]          [ idx 4 ] { addr: 0x0000000087f4e7f8, as_ptr: 0x0000000087f4e600, as_obj: { tp: -2013993472, ref: 0, reada
119  [SLAB]          [ idx 5 ] { addr: 0x0000000087f4e9f0, as_ptr: 0x0000000087f4e408, as_obj: { tp: -2013993976, ref: 0, reada
120  [SLAB]          [ idx 6 ] { addr: 0x0000000087f4ebe8, as_ptr: 0x0000000087f4e9f0, as_obj: { tp: -2013992464, ref: 0, reada
121  [SLAB]          [ idx 7 ] { addr: 0x0000000087f4ede0, as_ptr: 0x0000000087f4ebe8, as_obj: { tp: -2013991960, ref: 0, reada
122  [SLAB]      [ slab 0x0000000087eaa000 ] { freelist: 0x0000000087eaa9f0, nxt: 0x0000000000000000 }
123  [SLAB]          [ idx 0 ] { addr: 0x0000000087eaa018, as_ptr: 0x0000000087eaabe8, as_obj: { tp: -2014663704, ref: 0, reada
124  [SLAB]          [ idx 1 ] { addr: 0x0000000087eaa210, as_ptr: 0x0000000087eaa018, as_obj: { tp: -2014666728, ref: 0, reada
125  [SLAB]          [ idx 2 ] { addr: 0x0000000087eaa408, as_ptr: 0x0000000087eaa210, as_obj: { tp: -2014666224, ref: 0, reada
126  [SLAB]          [ idx 3 ] { addr: 0x0000000087eaa600, as_ptr: 0x0000000087eaa408, as_obj: { tp: -2014665720, ref: 0, reada
127  [SLAB]          [ idx 4 ] { addr: 0x0000000087eaa7f8, as_ptr: 0x0000000087eaa600, as_obj: { tp: -2014665216, ref: 0, reada
128  [SLAB]          [ idx 5 ] { addr: 0x0000000087eaa9f0, as_ptr: 0x0000000087eaa7f8, as_obj: { tp: -2014664712, ref: 0, reada
129  [SLAB]          [ idx 6 ] { addr: 0x0000000087eaabe8, as_ptr: 0x0000000087eaade0, as_obj: { tp: -2014663200, ref: 0, reada
130  [SLAB]          [ idx 7 ] { addr: 0x0000000087eaade0, as_ptr: 0x0000000000000000, as_obj: { tp: 0, ref: 0, readable: 5, wr
131  [My] Count matches
132  [SLAB] print kmem_cache end
```

此外，如果將空的slab忽略不印，則沒有報錯。

Reply

(http

林秉翰 (LIN, BING-HAN) (https://cool.ntu.edu.tw/courses/46296/users/188744)

1:14am

[更] 實作Bonus internal fragmentation issue 後，不會報錯，輸出如下。針對上述[Invalid print message:{.....}is broken or not in partial]**具體原因仍不明**。

```
125  $ mp2
126  [SLAB] kmem_cache { name: file_cache, object_size: 504, at: 0x0000000087f59000, in_cache_obj: 7 }
127  [SLAB]  [ cache slabs ]
128  [SLAB]      [ slab 0x0000000087f59000 ] { freelist: 0x0000000087f59260, nxt: 0x0000000000000000 }
129  [SLAB]          [ idx 0 ] { addr: 0x0000000087f59068, as_ptr: 0x0000000900000003, as_obj: { tp: 3, ref: 9, readable: 1, wr
130  [SLAB]          [ idx 1 ] { addr: 0x0000000087f59260, as_ptr: 0x0000000087f59c38, as_obj: { tp: -2013946824, ref: 0, reada
131  [SLAB]          [ idx 2 ] { addr: 0x0000000087f59458, as_ptr: 0x0000000087f59848, as_obj: { tp: -2013947832, ref: 0, reada
132  [SLAB]          [ idx 3 ] { addr: 0x0000000087f59650, as_ptr: 0x0000000087f59a40, as_obj: { tp: -2013947328, ref: 0, reada
133  [SLAB]          [ idx 4 ] { addr: 0x0000000087f59848, as_ptr: 0x0000000000000000, as_obj: { tp: 0, ref: 0, readable: 0, wr
134  [SLAB]          [ idx 5 ] { addr: 0x0000000087f59a40, as_ptr: 0x0000000087f59458, as_obj: { tp: -2013948840, ref: 0, reada
135  [SLAB]          [ idx 6 ] { addr: 0x0000000087f59c38, as_ptr: 0x0000000087f59650, as_obj: { tp: -2013948336, ref: 0, reada
136  [SLAB]  [ partial slabs ]
137  [SLAB]      [ slab 0x0000000087eab000 ] { freelist: 0x0000000087eabbe8, nxt: 0x0000000087f59030 }
138  [SLAB]          [ idx 0 ] { addr: 0x0000000087eab018, as_ptr: 0x0000000087eabde0, as_obj: { tp: -2014659104, ref: 0, reada
139  [SLAB]          [ idx 1 ] { addr: 0x0000000087eab210, as_ptr: 0x0000000087eab018, as_obj: { tp: -2014662632, ref: 0, reada
140  [SLAB]          [ idx 2 ] { addr: 0x0000000087eab408, as_ptr: 0x0000000087eab210, as_obj: { tp: -2014662128, ref: 0, reada
141  [SLAB]          [ idx 3 ] { addr: 0x0000000087eab600, as_ptr: 0x0000000087eab408, as_obj: { tp: -2014661624, ref: 0, reada
142  [SLAB]          [ idx 4 ] { addr: 0x0000000087eab7f8, as_ptr: 0x0000000087eab600, as_obj: { tp: -2014661120, ref: 0, reada
143  [SLAB]          [ idx 5 ] { addr: 0x0000000087eab9f0, as_ptr: 0x0000000087eab7f8, as_obj: { tp: -2014660616, ref: 0, reada
144  [SLAB]          [ idx 6 ] { addr: 0x0000000087eabbe8, as_ptr: 0x0000000087eab9f0, as_obj: { tp: -2014660112, ref: 0, reada
145  [SLAB]          [ idx 7 ] { addr: 0x0000000087eabde0, as_ptr: 0x0000000000000000, as_obj: { tp: 0, ref: 0, readable: 5, wr
```

Edited by 林秉翰 (LIN, BING-HAN) (https://cool.ntu.edu.tw/courses/46296/users/188744) on Apr 3 at 1:15am

Reply

(http

楊子慶 (YANG, TZU-CHING) (https://cool.ntu.edu.tw/courses/46296/users/229158)

4:11am

同學您好：

感謝同學的追查與回報。

助教審視測試系統後發現確實在 3/30 的提交 (https://github.com/Shiritai/xv6-ntu-mp2/commit/b824be1a8e462606d2c9211c4d0945c8881e1475) 中，新增對 partial 的檢查遺漏考慮 free 與 partial 置於同一串列的部分，對此進行修正可見於本提交 (https://github.com/Shiritai/xv6-ntu-mp2/commit/31a9390b2097ac3432a7a667be74a21d820222bf)，很抱歉由於助教的疏失造成同學的困擾。

本次修正提交前助教已經針對 8 種實作（使用 full/free/in-cache 與否）進行完整的測試，目前沒有遇到問題。

若有更多問題請不吝提出，感謝。

子慶

Reply



林秉翰 (LIN, BING-HAN) (https://cool.ntu.edu.tw/courses/46296/users/188744)

3:52pm

感謝助教百忙中回覆及修正。

Reply